

APMA 4302: Methods in Computational Science

Spring 2026

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Lecture 19

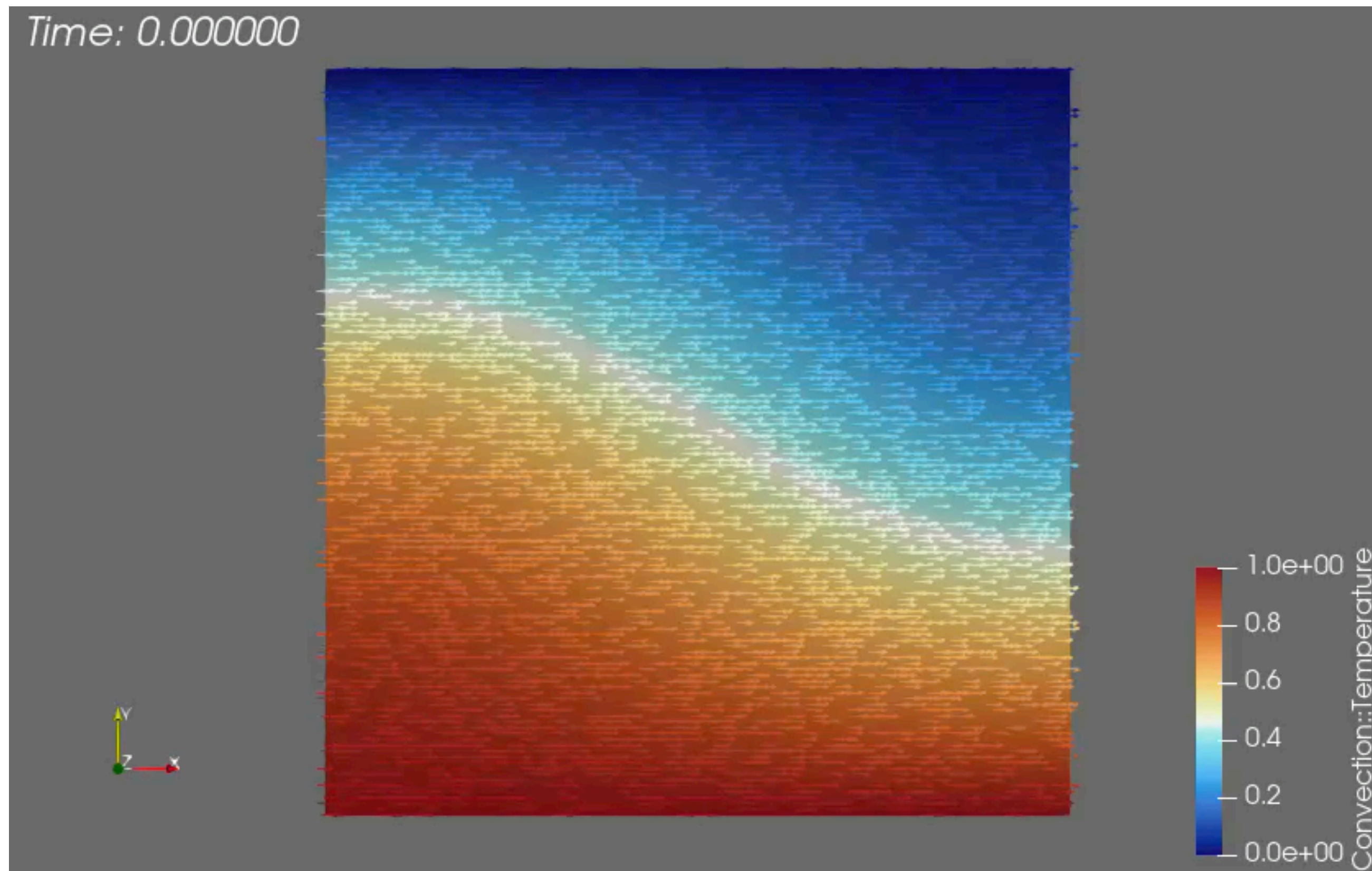
Advection-Diffusion Problems (Beuler Ch 11)

Outline

- Motivation and Challenges
- Basic physics: The Method of characteristics
- Finite Volume discretization
- Pure advection: 1-D and 2-D
- Advection + Diffusion: Arkimex schemes

Multi-physics problems

Towards effective multi-physics models



Advection-Diffusion Eq.

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \frac{1}{Ra} \nabla^2 T$$

$$-\nabla \cdot \eta [(\nabla \mathbf{v}) + (\nabla \mathbf{v})^T] + \nabla P = T \hat{\mathbf{g}}$$

$$\nabla \cdot \mathbf{v} = 0$$

where $Ra = \frac{\rho_0 \alpha \Delta T h^3}{\eta \kappa}$ is the Raleigh #

and $\eta = \eta(T, \mathbf{v})$ is the viscosity

Incompressible Stokes flow

Problem #1: Infinite Prandtl number thermal convection

Other problems

Advection dominated multi-physics problems

- Navier Stokes/CFD
- Reactive Flow and Transport
- MHD/Plasma Physics
- Geophysical Fluid Dynamics
- Combustion
- Astrophysical fluid dynamics

Conservative and non-conservative forms

Advection-Diffusion Reaction Equations

Conservative form

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{F} = g(u, t, \mathbf{x})$$

with Flux

$$\mathbf{F} = \mathbf{v}u - \nu \nabla u + \dots$$

Non-conservative form

$$\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = \nabla \cdot \nu \nabla u + f(u, t, \mathbf{x})$$

pure advection

Solution by method of characteristics

$$\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = 0$$

Is actually equivalent to a system of ODE's on particles

pure advection

Themes and variations: Source terms, non-linear problems

$$\frac{\partial u}{\partial t} + \mathbf{v} \cdot \nabla u = g(u, t, \mathbf{x})$$

can solve for $\mathbf{v}(\mathbf{x})$ or $\mathbf{v}(\mathbf{x}, u)$

pure advection

Finite Volume methods

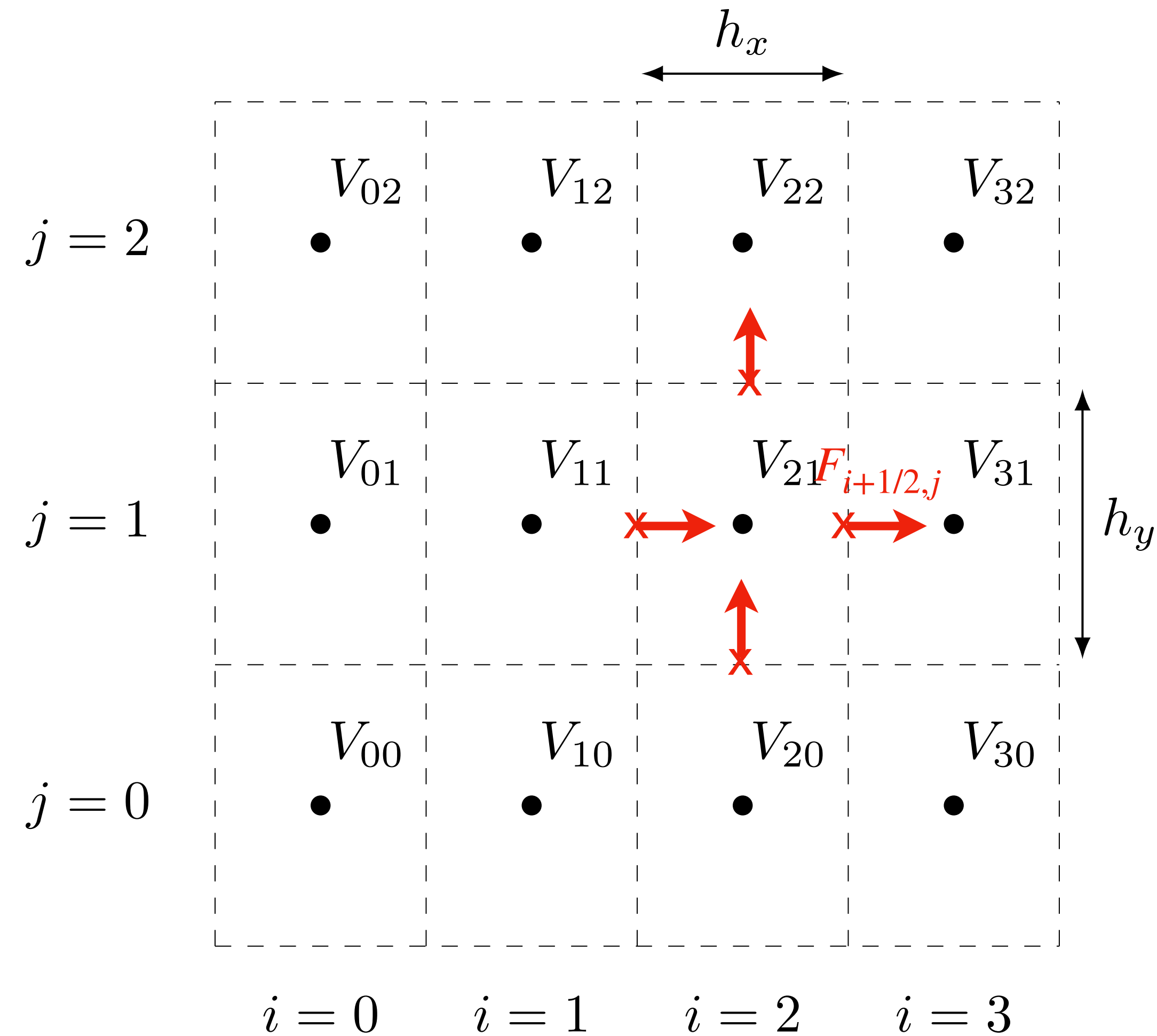
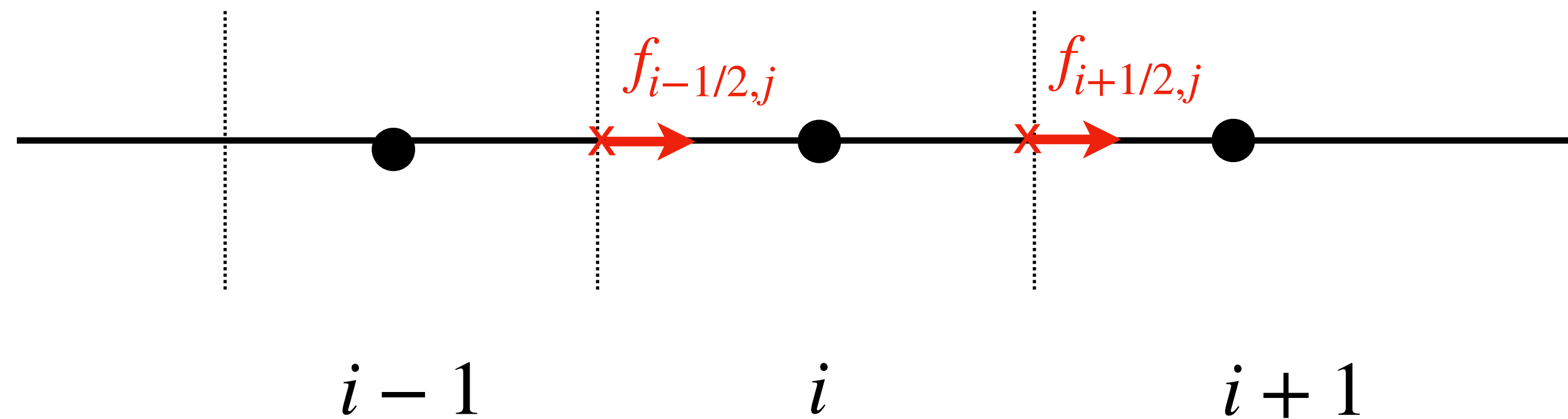


Figure 11.1. Control volumes V_{ij} with cell centers at (x_i, y_j) .

pure advection

Finite Volume methods 1-D, advection

$$\frac{\partial u}{\partial t} + \frac{\partial f}{\partial x} = 0$$

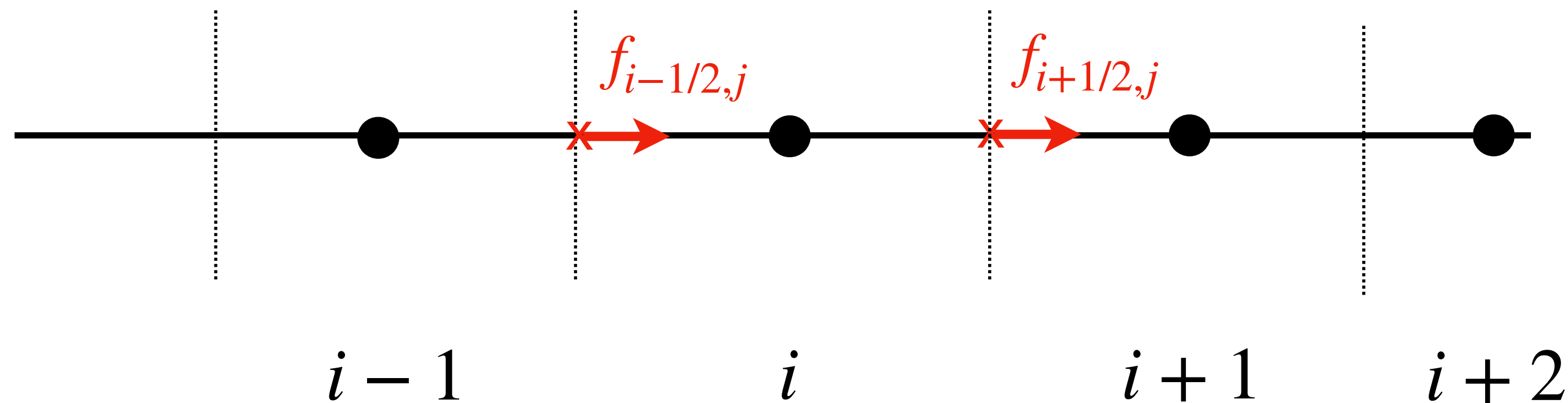


$$f = au$$

$$\frac{\partial u}{\partial t} + \frac{\partial au}{\partial x} = 0$$

pure advection

Finite Volume methods 1-D advection, choices for f



solve MOL system
$$\dot{U}_i + \frac{f_{i+1/2} - f_{i-1/2}}{h} = 0$$

centered:
$$f_{i+1/2} = a(x_{i+1/2})(U_{i+1} + U_i)/2$$

upwind:
$$f_{i+1/2} = a^+(x_{i+1/2})U_i + a^-(x_{i+1/2})U_{i+1}$$

where:
$$a^+ = \frac{a + |a|}{2} = \max(0, a) \quad a^- = \frac{a - |a|}{2} = \min(0, a)$$

Demonstration:

New PETSc code `ts_advection_diffusion` (in methods repo) solves

$$\frac{\partial u}{\partial t} + \frac{\partial au}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

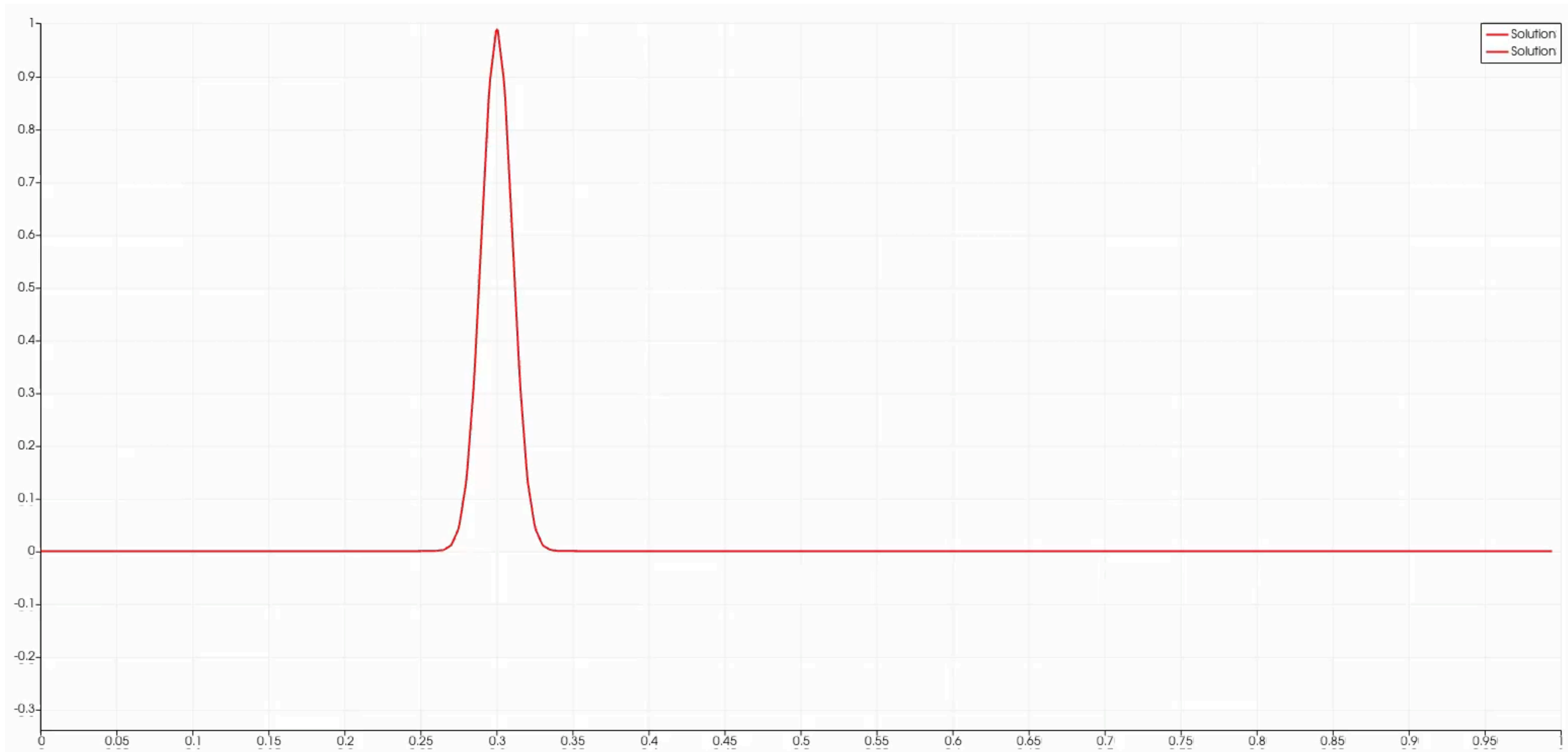
using TS (with arkimex solvers), for constant a and analytic solution for a diffusing translating Gaussian

$$u(x, t) = \frac{\sigma}{\sqrt{\sigma^2 + 2\nu t}} \exp\left(-\frac{(x - x_0 - at)^2}{2(\sigma^2 + 2\nu t)}\right).$$

for different choices for f . for $\nu = 0$ is just pure advection

Comparison upwind v centered:

Poorly resolved initial condition $\sigma = 0.01$ $\nu = 0$ (pure advection)



pure advection: issues

MOL methods become the linear ODE's $\frac{d\mathbf{U}}{dt} = A\mathbf{U}$ with Different A's

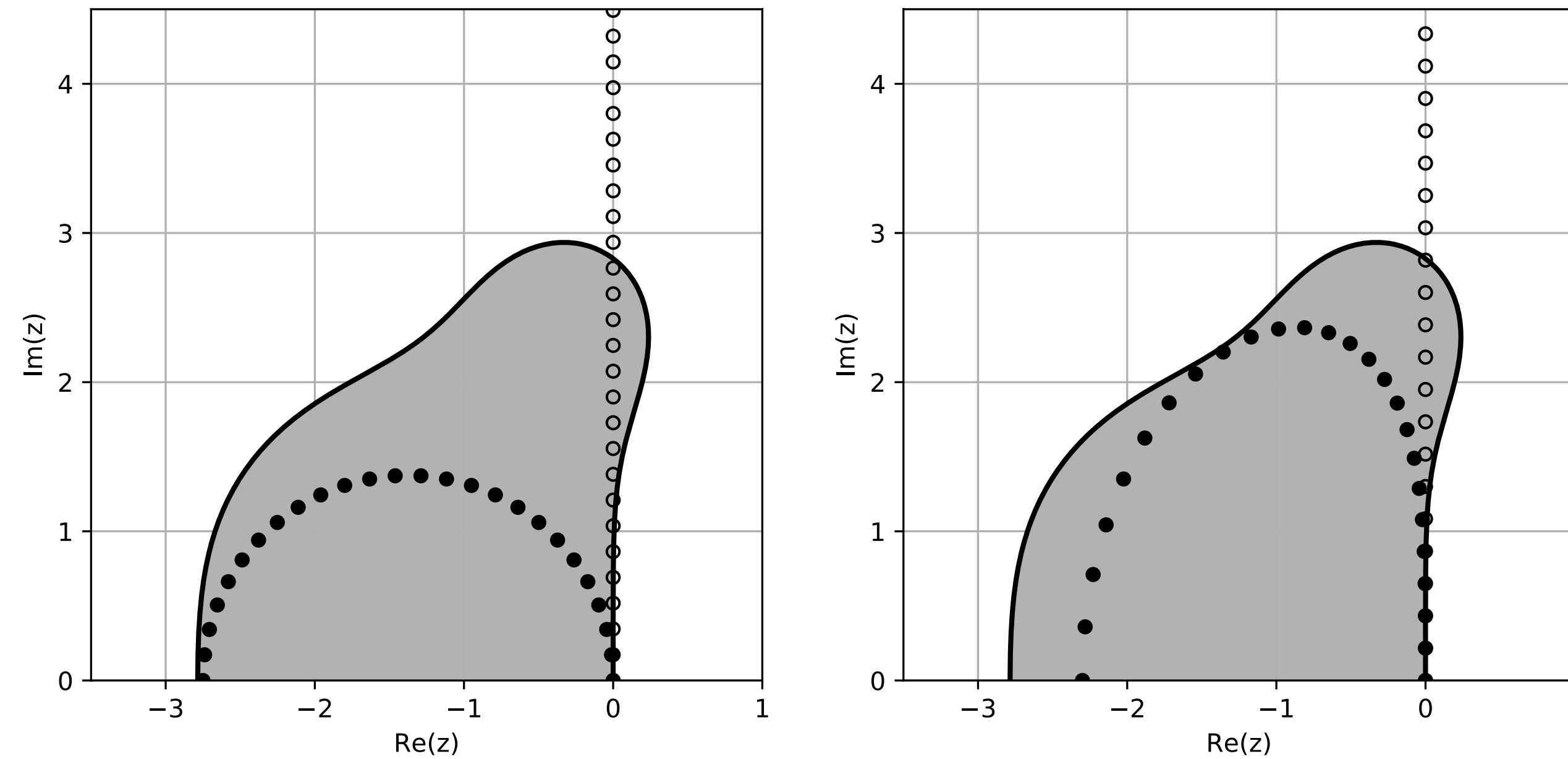
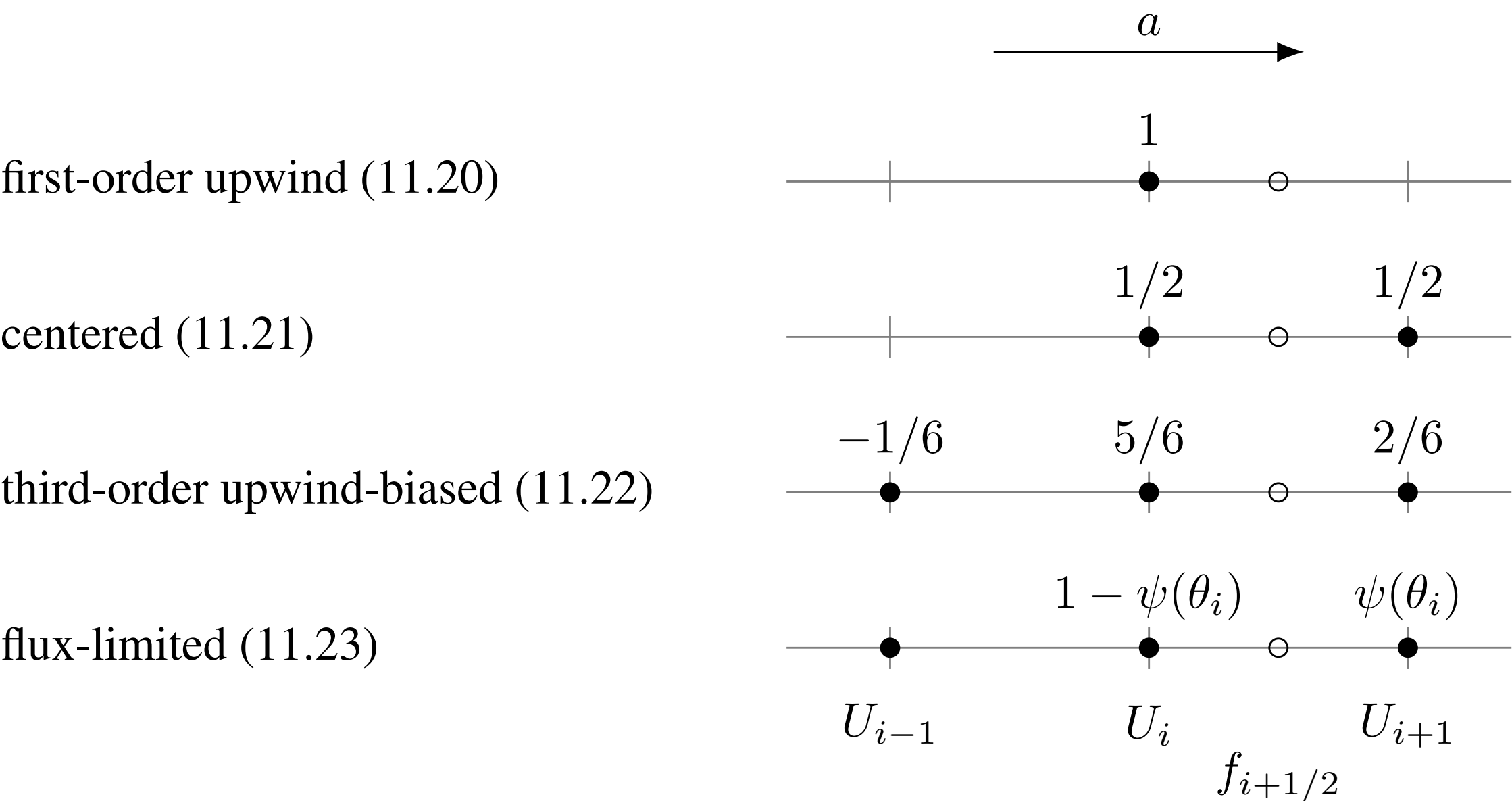


Figure 11.3. The exact eigenvalues of PDE (11.11) (open circles) are imaginary. Those from first-order upwinding on the MOL system (11.12) (left; solid) or a third-order upwind-biased scheme (11.22) (right; solid) are damped by negative real parts. The absolute stability region of RK4 is shaded.

Fixes (?):



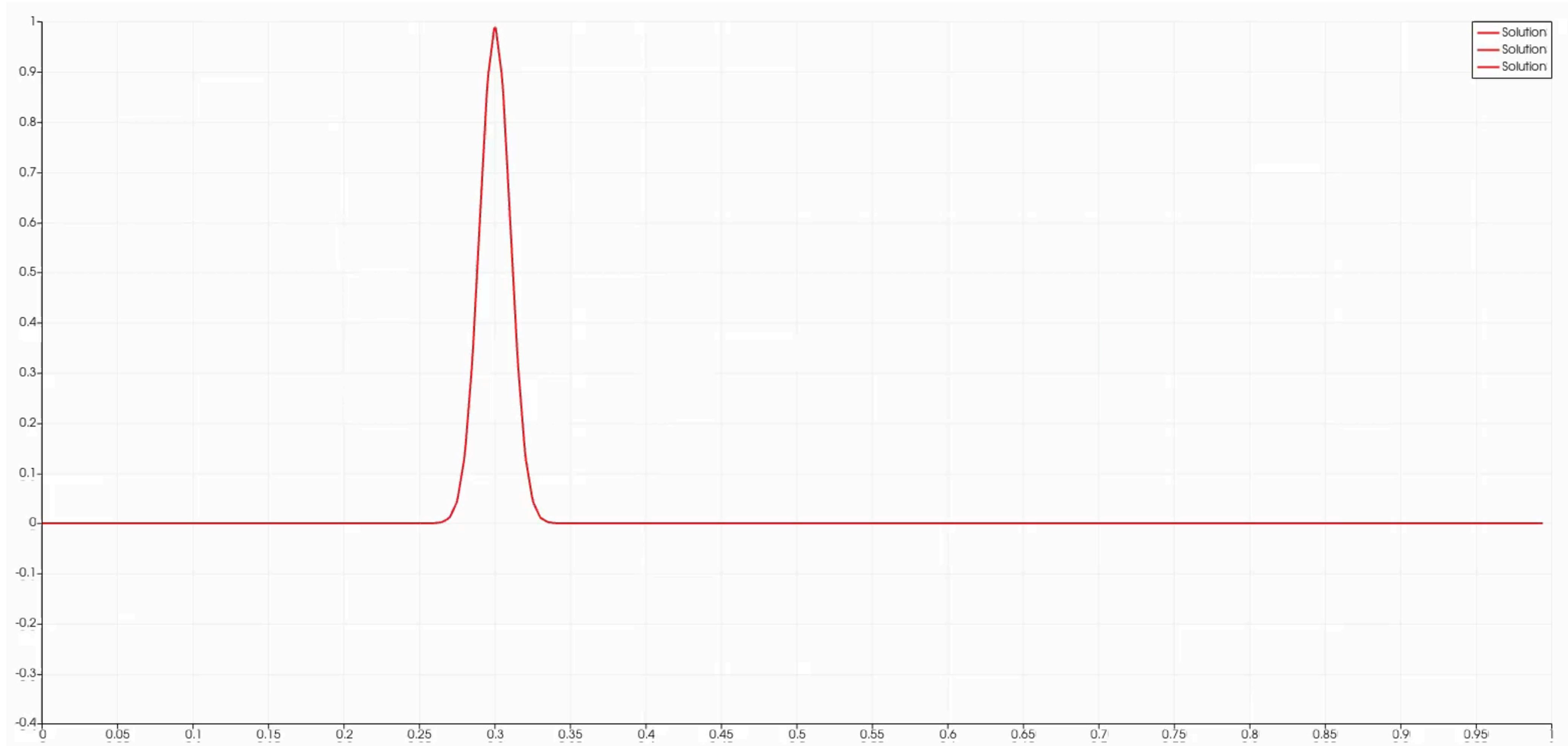
$$\theta_i = \frac{U_i - U_{i-1}}{U_{i+1} - U_i} \quad \text{ratio of slopes}$$

Figure 11.5. When $a = a(x_{i+1/2}) \geq 0$, schemes (11.23) all use the same three regular grid points, with coefficients as shown.

2-D demo: Bueler Ch11 p4pde's advect.c

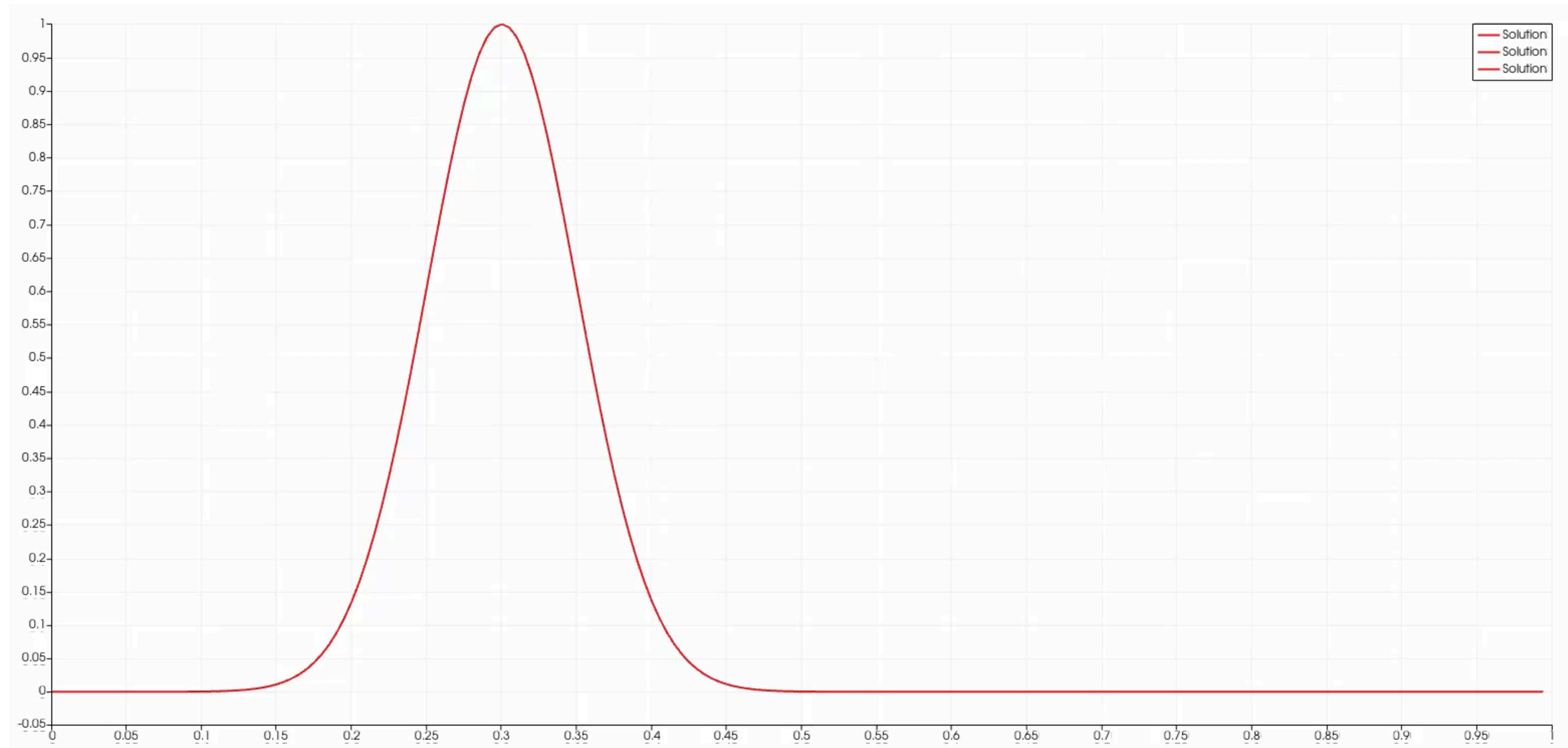
Comparison adding 3rd order biased upwind:

Poorly resolved initial condition $\sigma = 0.01$ $\nu = 0$ (pure advection)



Comparison adding 3rd order biased upwind:

well resolved initial condition $\sigma = 0.05$ $\nu = 0$ (pure advection)



Flux Limiters (2nd order TVD schemes)

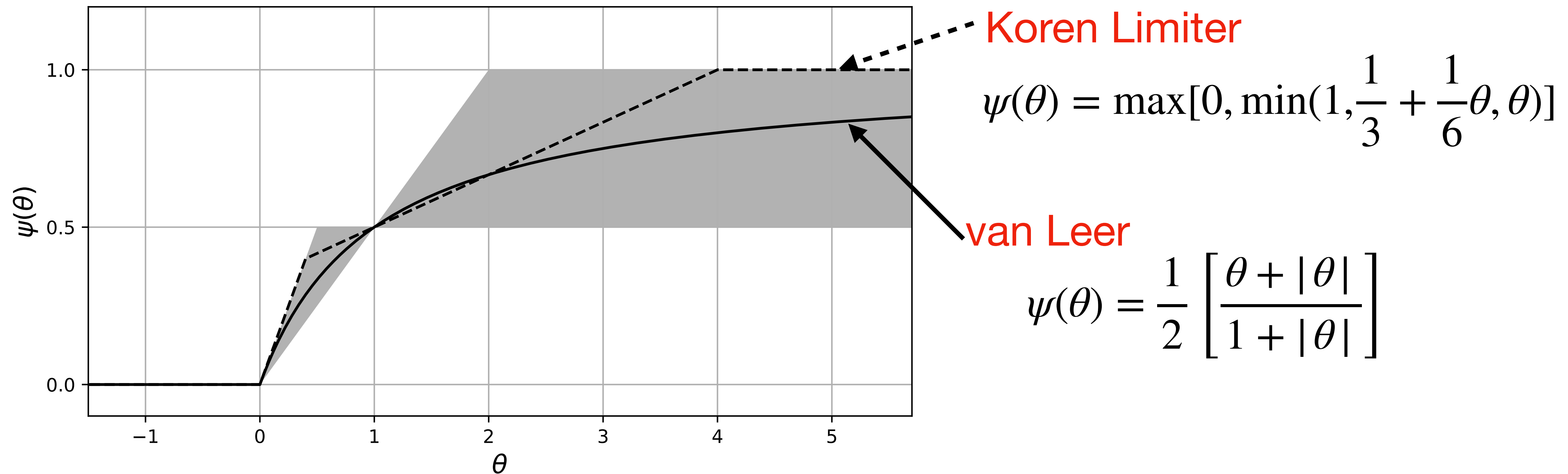
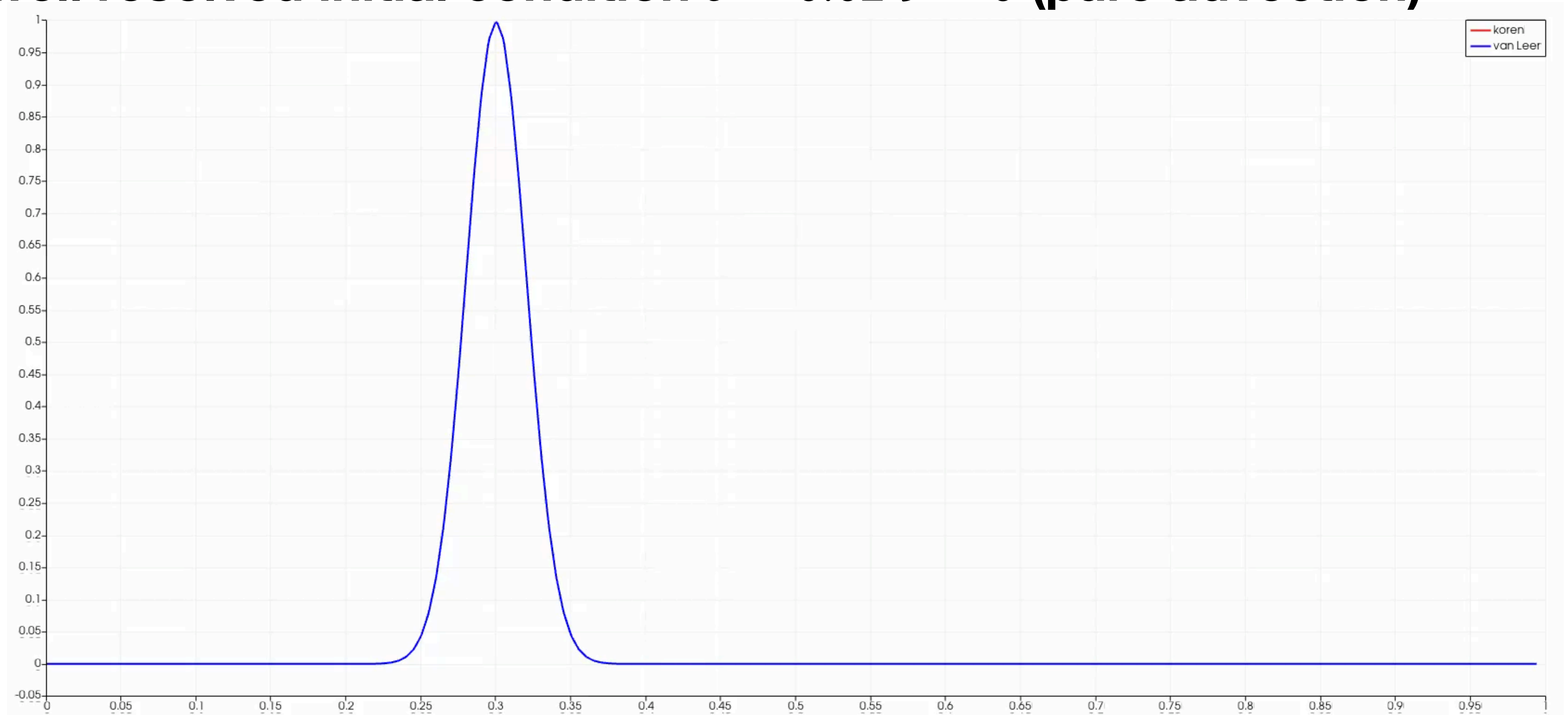


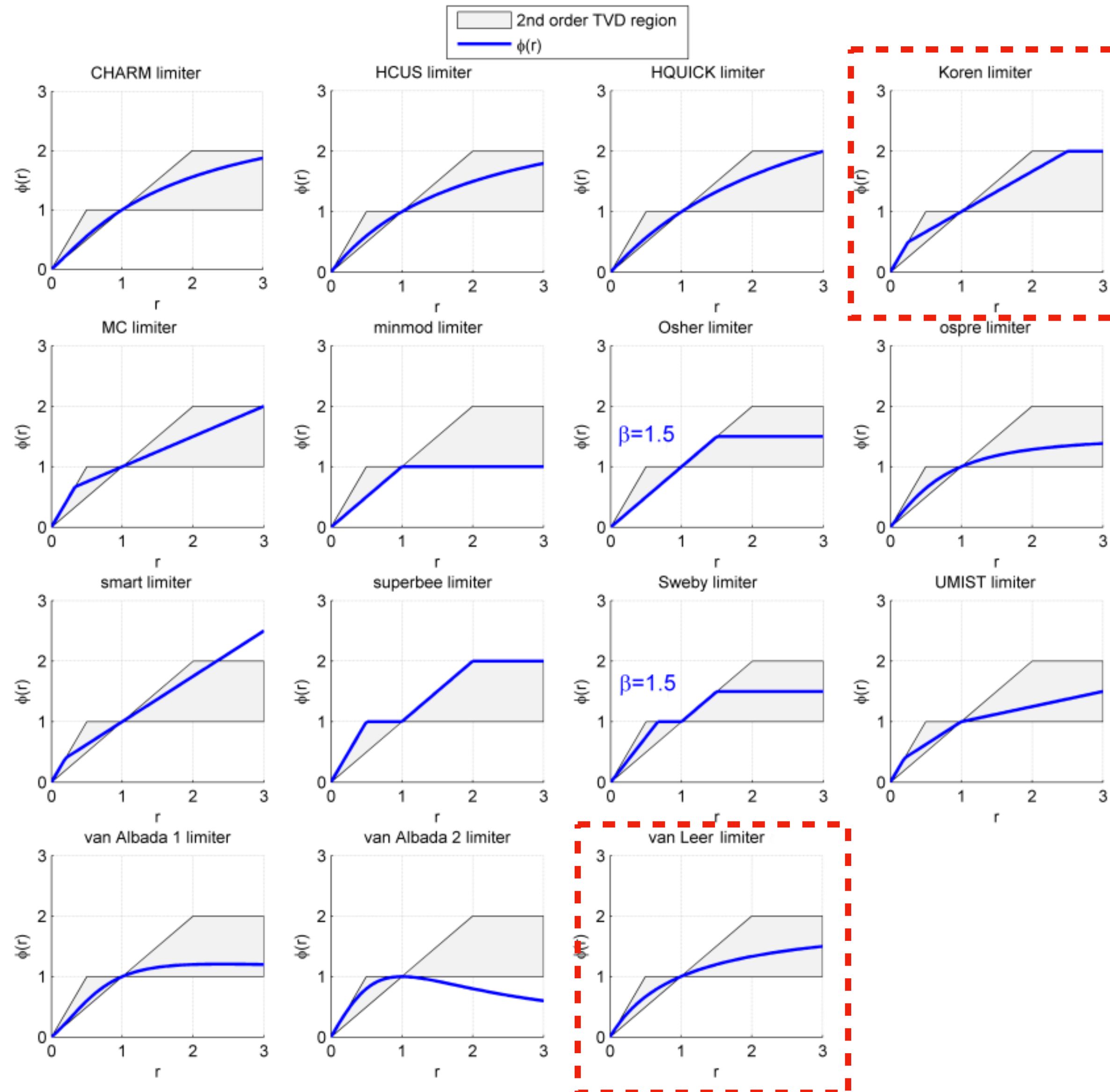
Figure 11.6. The Koren (11.25) (dashed) and van Leer (11.26) (solid) limiters are two of many possible flux-limiters $\psi(\theta)$. Flux-limiters give second-order results and avoid spurious oscillations if they lie in the shaded Sweby region [138].

Comparison koren v van Leer limiters:

well resolved initial condition $\sigma = 0.02$ $\nu = 0$ (pure advection)



A zoology of flux limiters

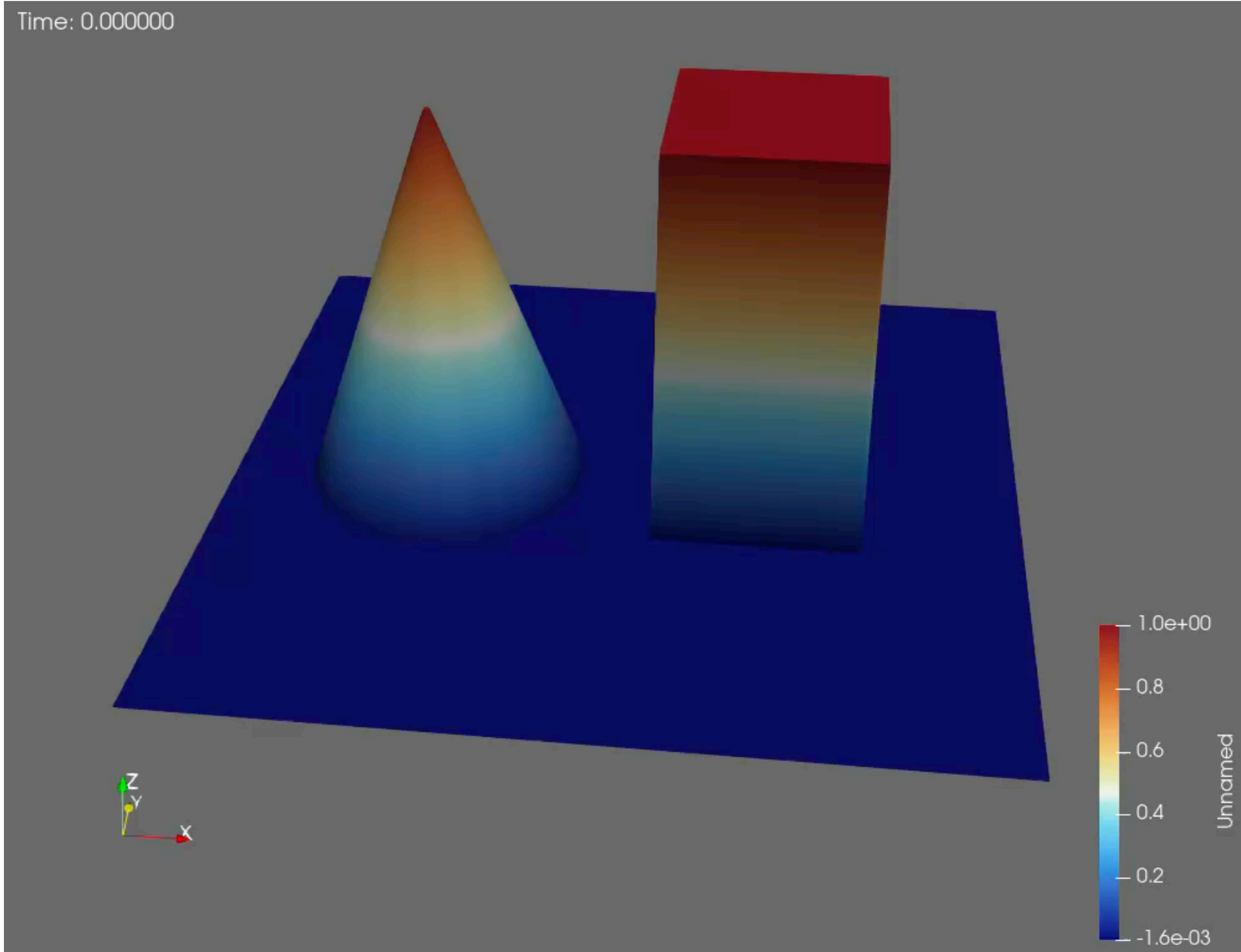
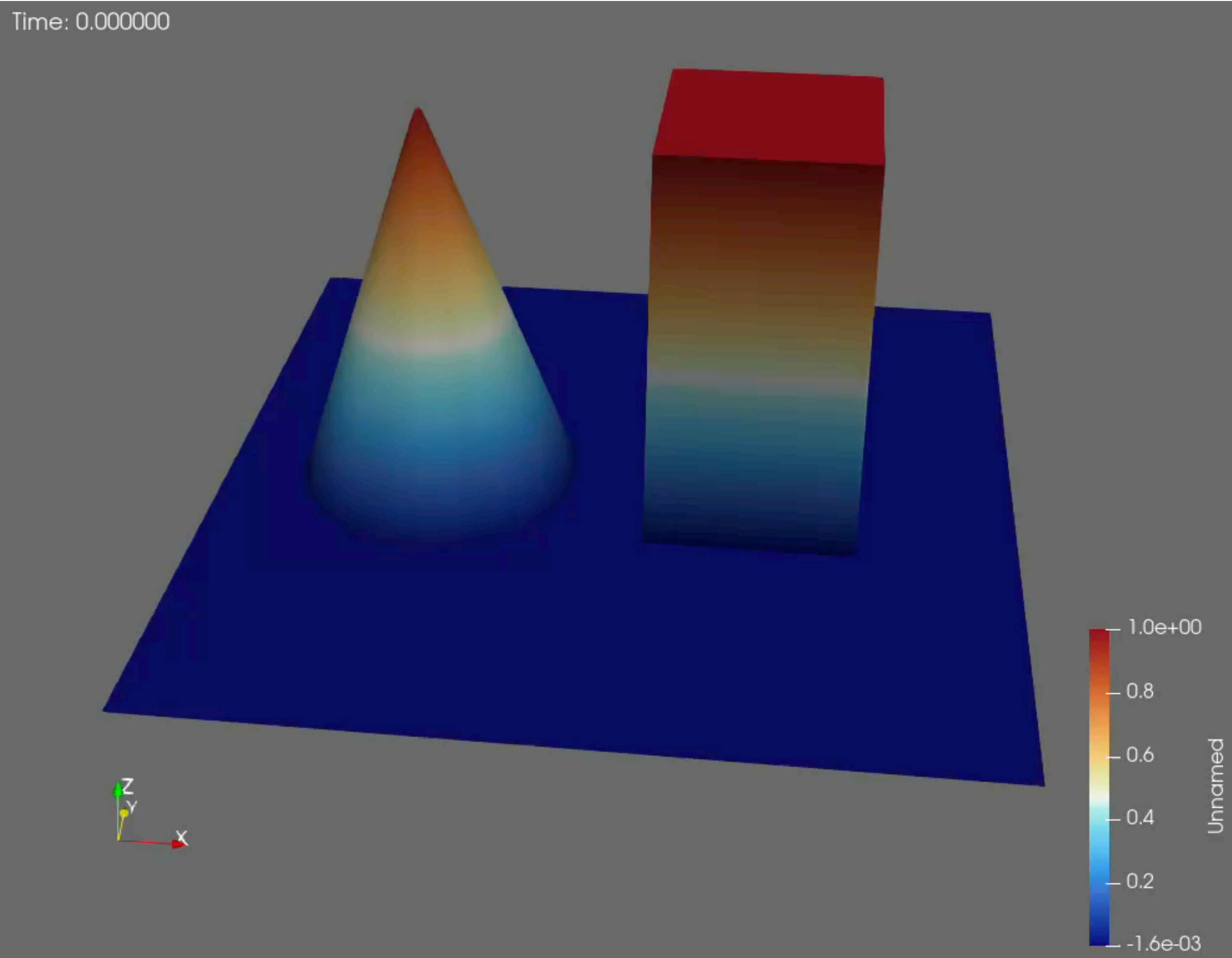


comparison of flux-limiters 2-D pure advection

Bueler advect.c

Koren limiter

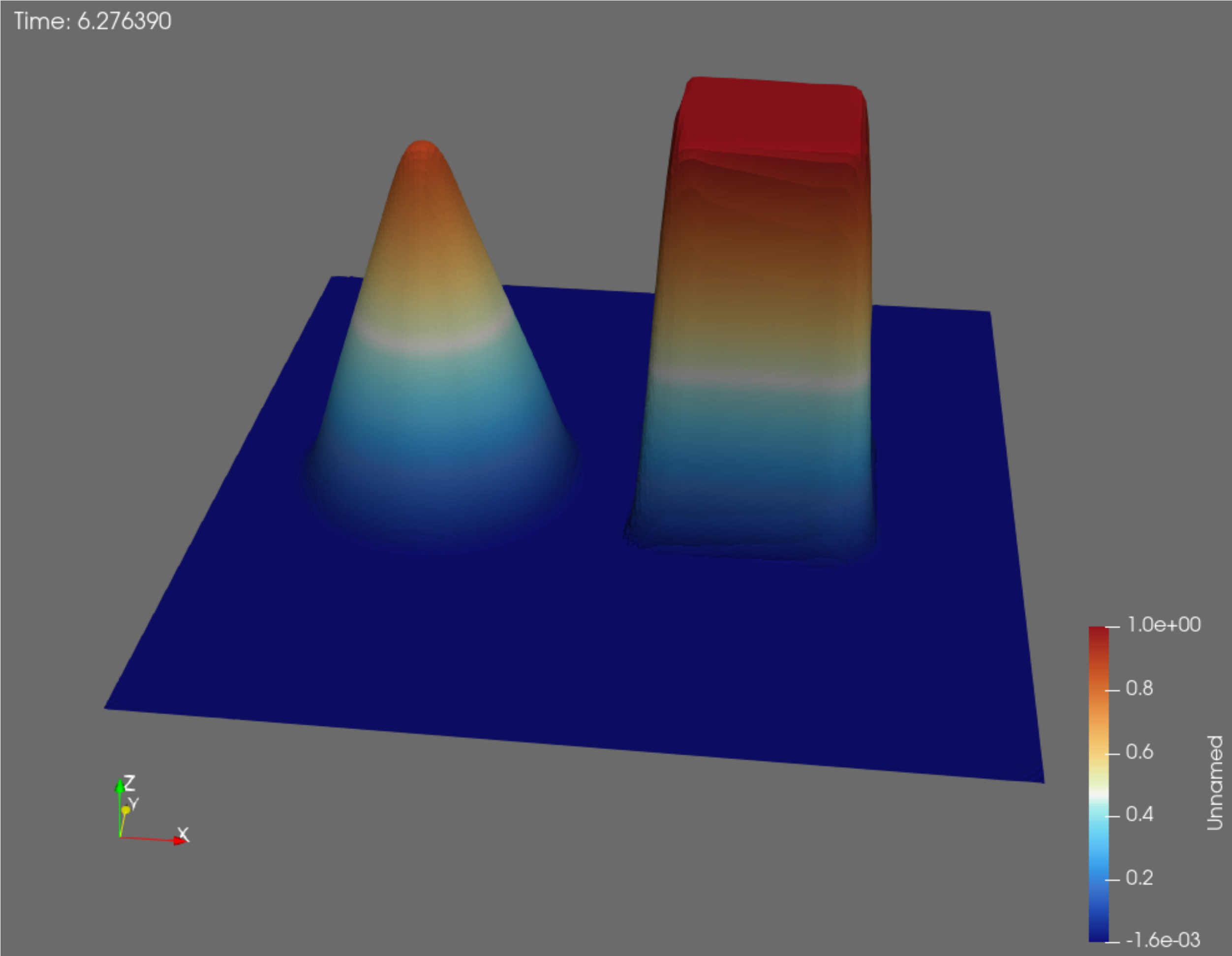
vanLeer limiter



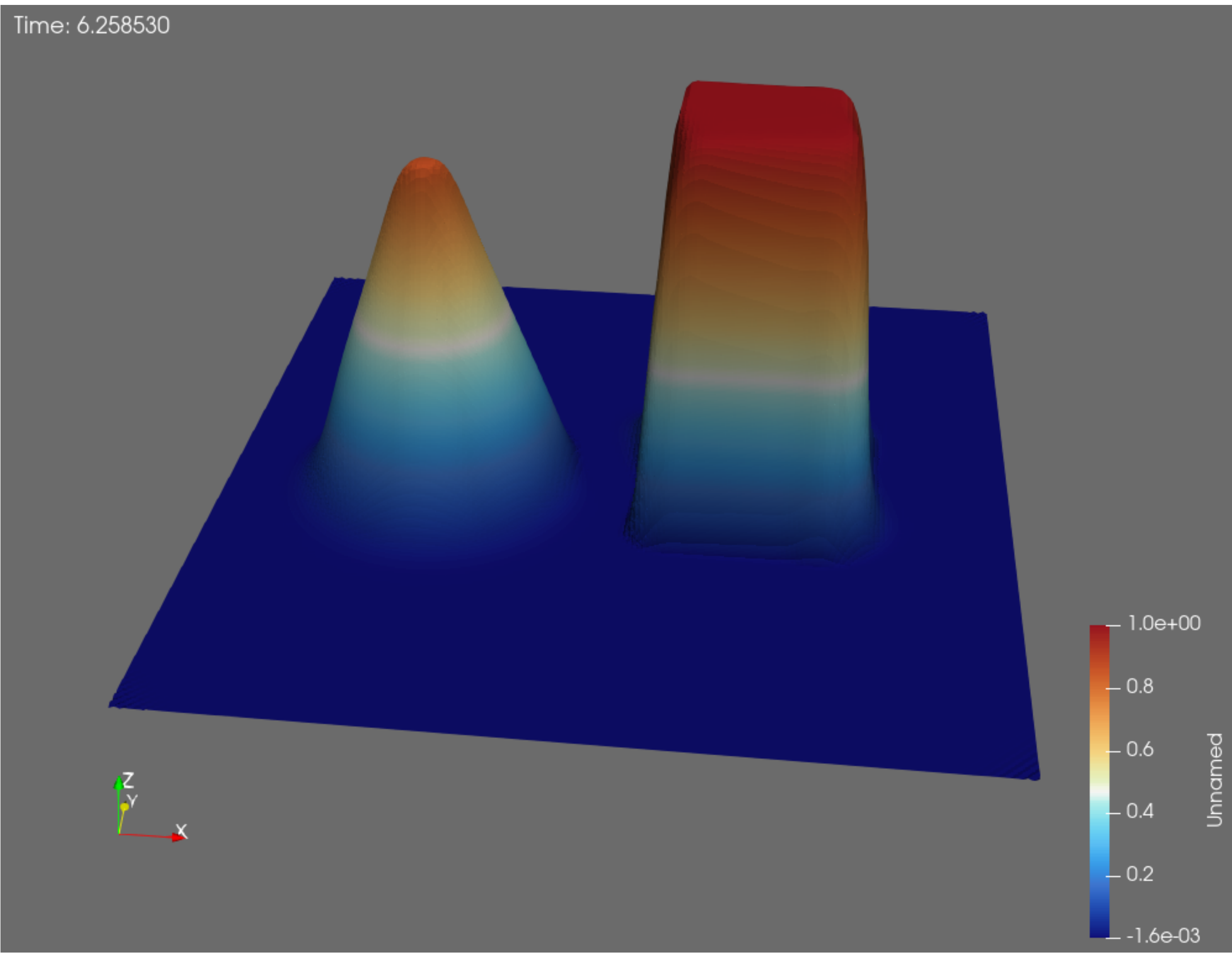
comparison of flux-limiters 2-D pure advection

Bueler advect.c

Koren limiter

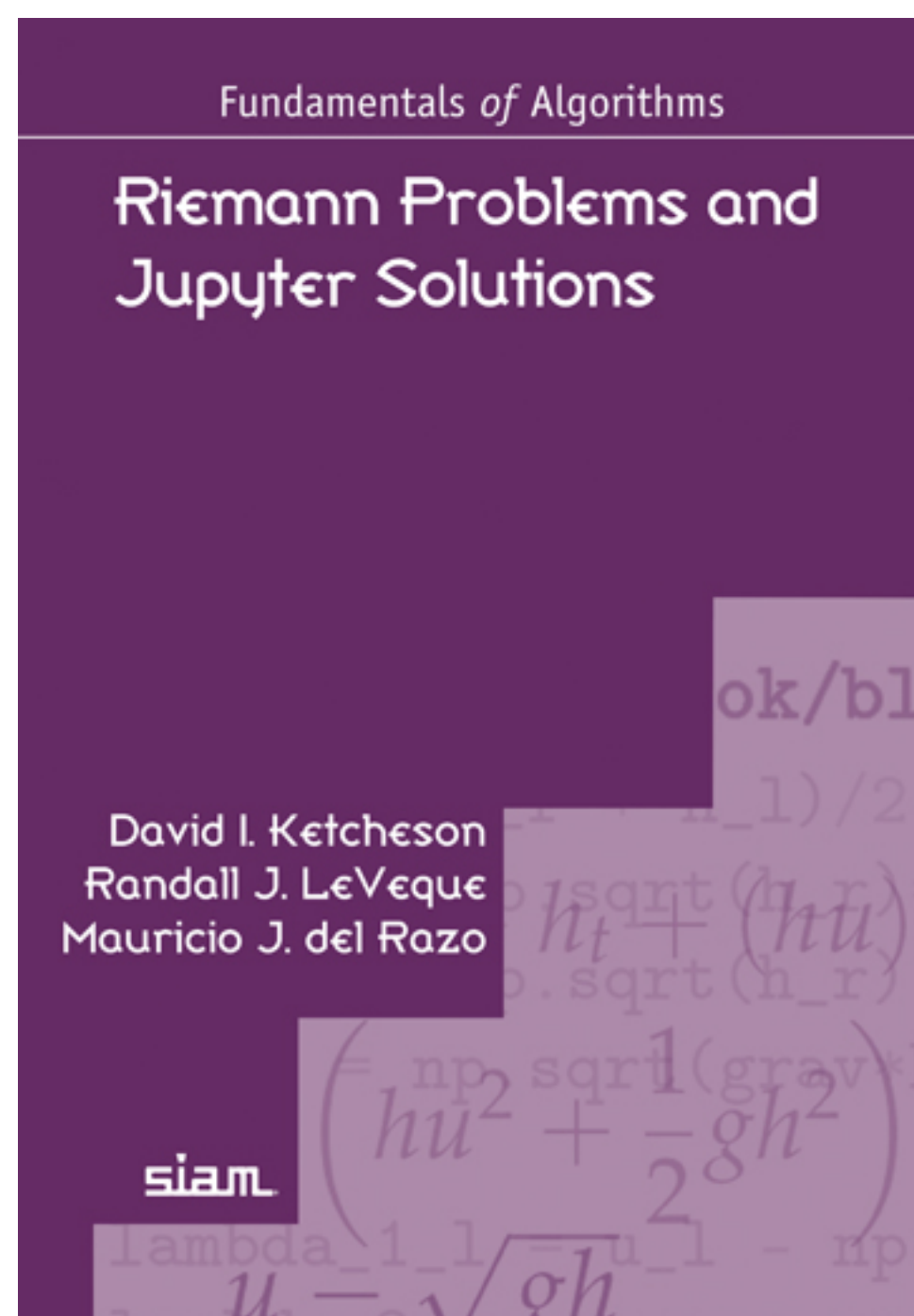


vanLeer limiter

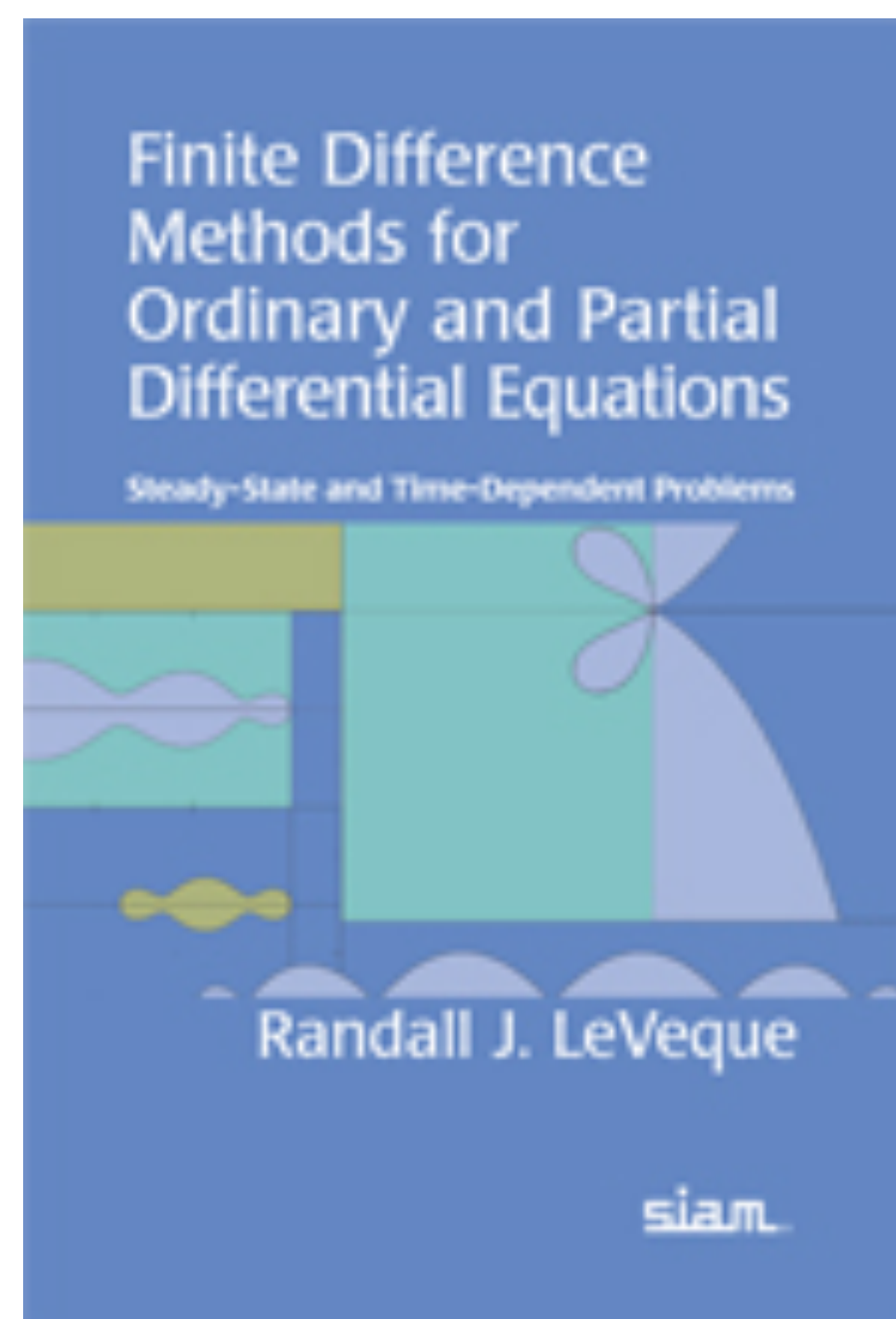


References for Hyperbolic systems and PDEs

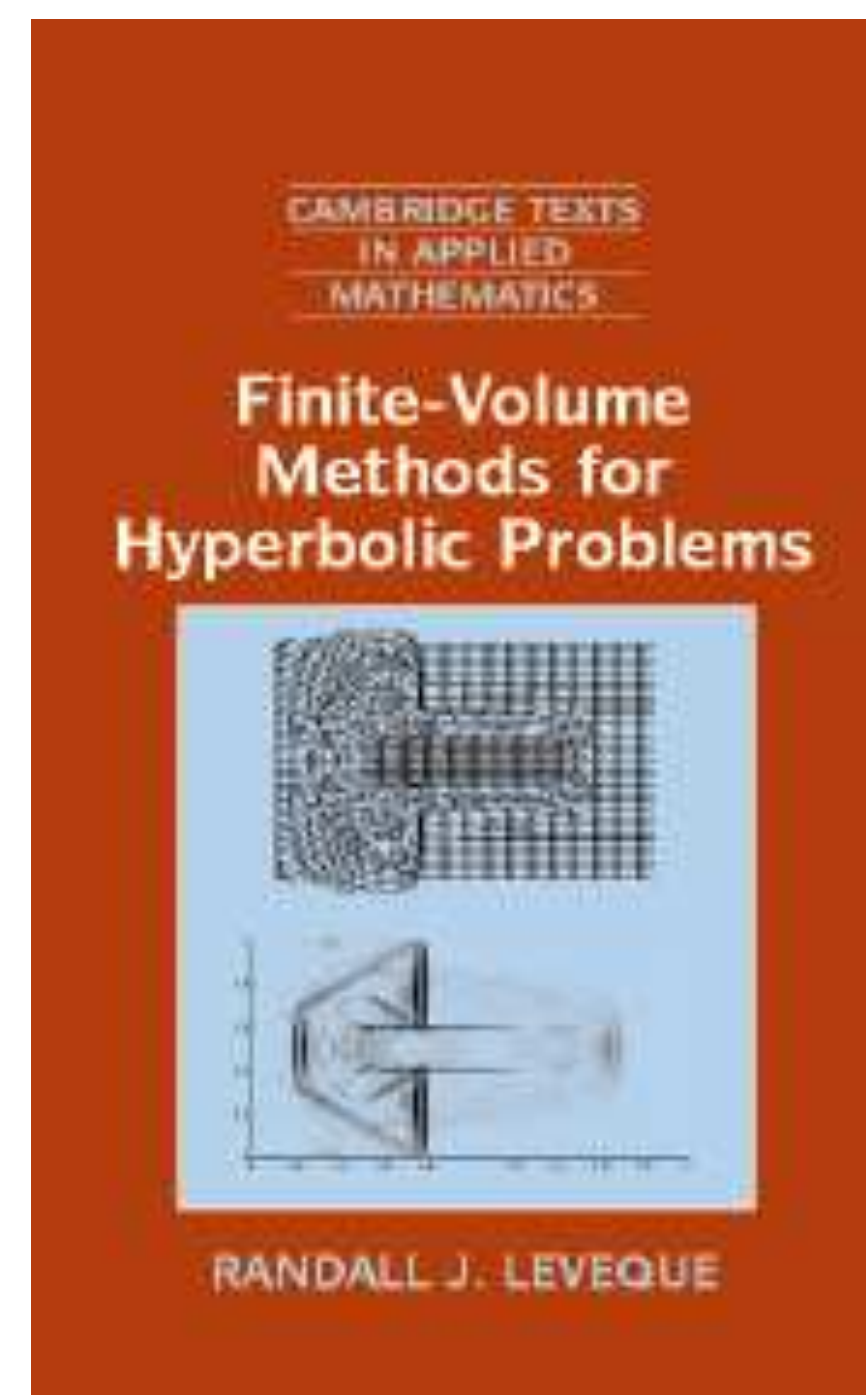
R. LeVeque <https://faculty.washington.edu/rjl/booksnotes.html>



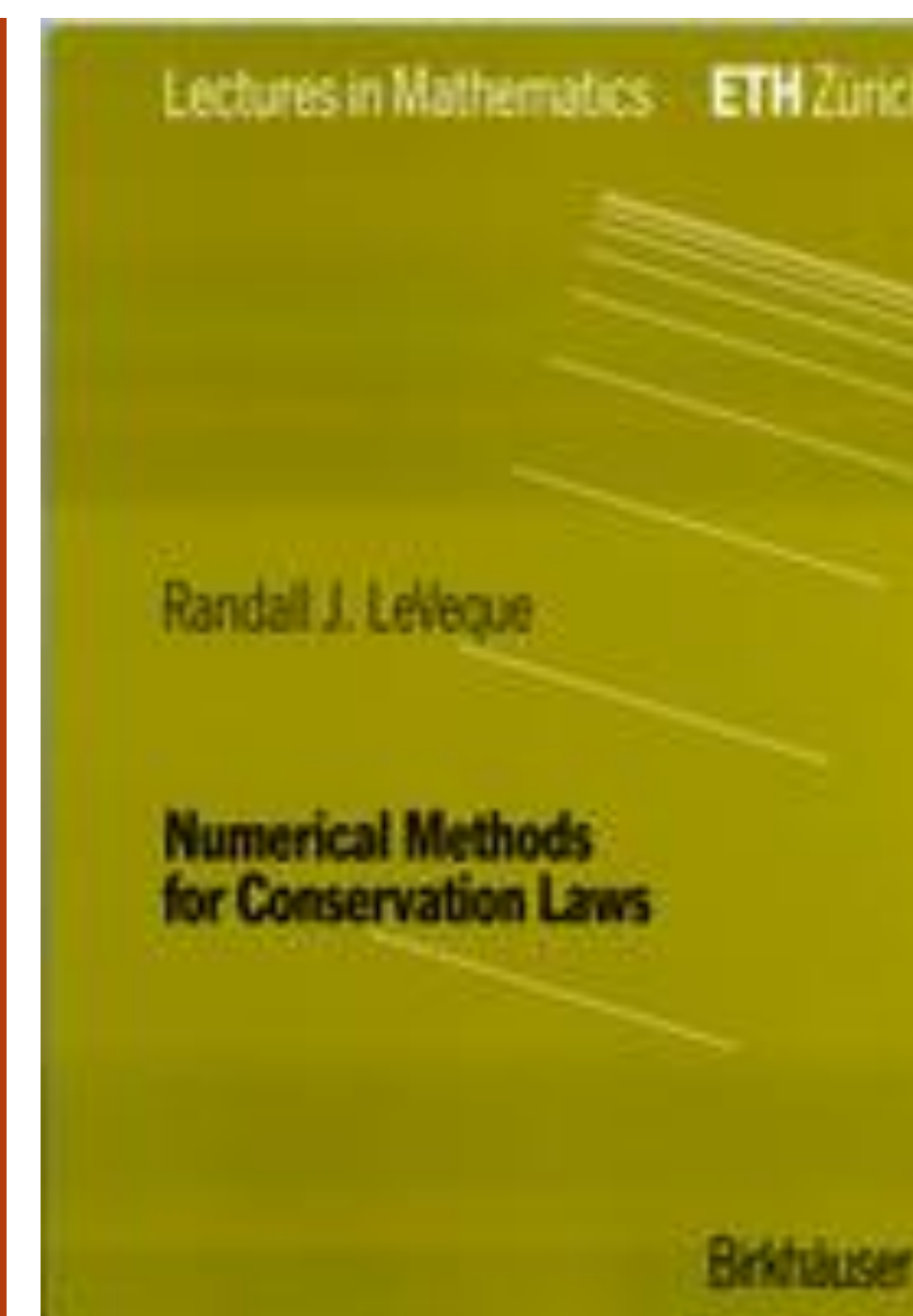
2020



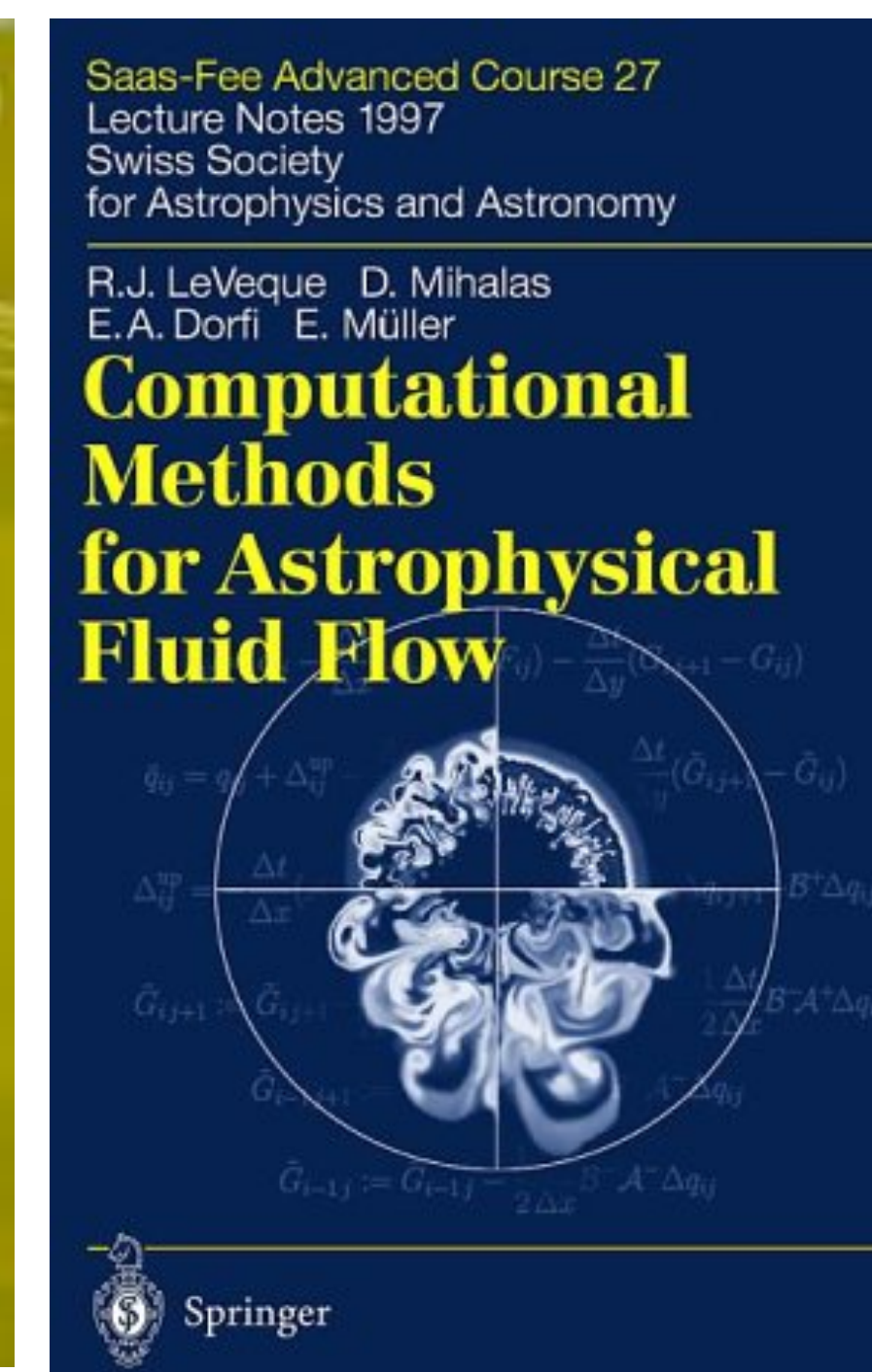
2007



2002



1990



1998

Combined Advection-Diffusion

TS Arkimex schemes

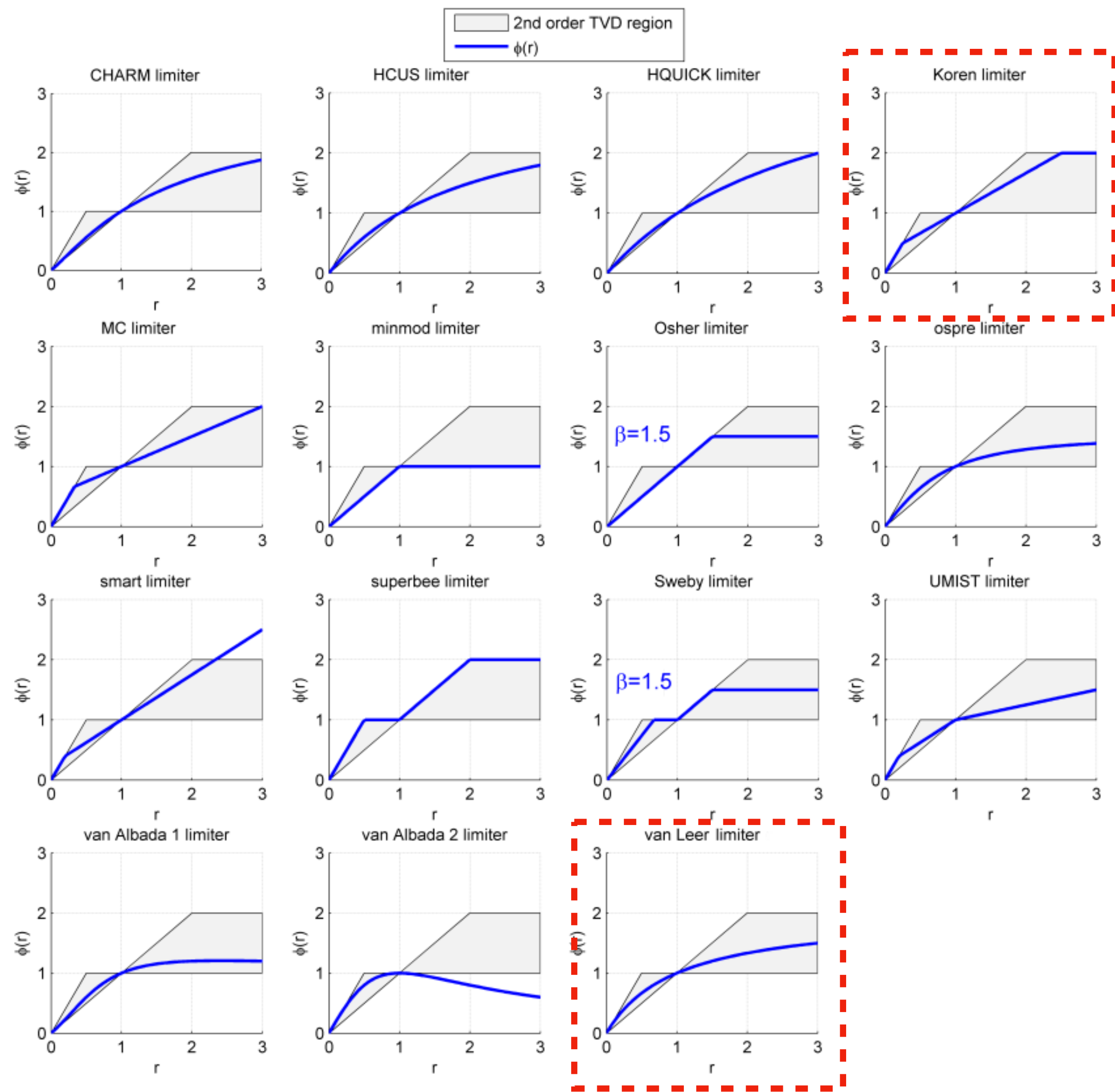
$$\frac{\partial u}{\partial t} + \frac{\partial au}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

Two separate processes with very different optimal solvers

- Diffusion: stiff works well with L-stable implicit solvers (BDF, BE)
- Advection: works well with explicit solvers (and poorly with implicit)

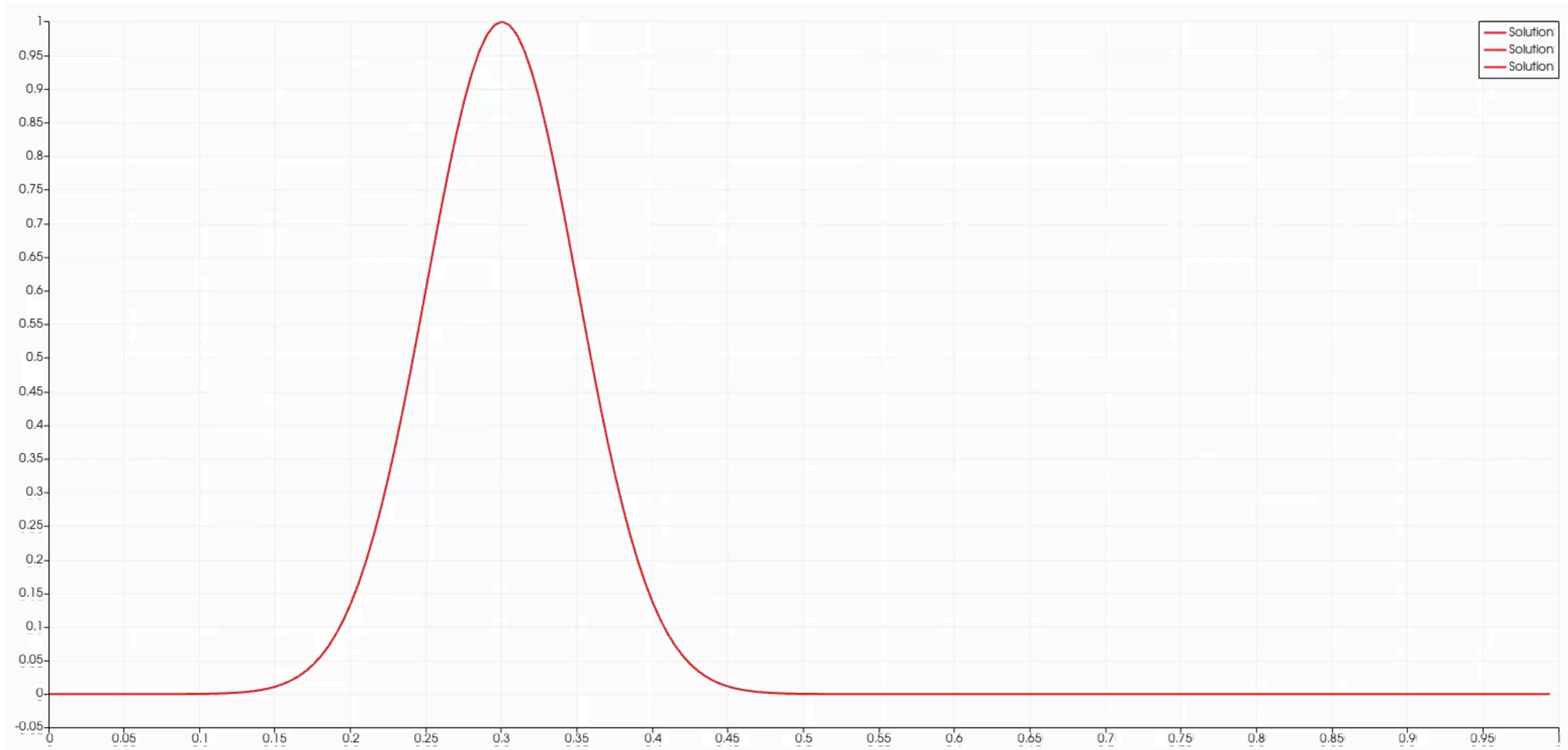
Table 11.3. Run times and L^1 errors for implicit (cn) and explicit (rk) time-stepping on a pure advection problem.

TS type	limiter	solver options	time (s)	L^1 error
cn	none	-adv_jac_limiter none	6.4	7.6×10^{-2}
	centered	-adv_jac_limiter centered	12.8	5.0×10^{-2}
	vanleer	-adv_jac_limiter none	67.6	1.0×10^{-2}
	vanleer	same + -snes_mf_operator	54.8	1.0×10^{-2}
rk	none		0.5	7.6×10^{-2}
	centered		0.5	3.7×10^{-2}
	vanleer		1.0	1.2×10^{-2}



Adding a little diffusion: arkimex

well resolved initial condition $\sigma = 0.05$ $\nu = 0.001$



Comparison adding 3rd order biased upwind:

well resolved initial condition $\sigma = 0.05$ $\nu = 0$ (pure advection)

